Fiscal Year:	FY 2009	**Task Last Updated:**	FY 10/07/2009
PI Name:	Musgrave, Mary Ph.D.		
Project Title:	Biological and Physical Constraints on Seed Development in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Zip Code:	06269	**Congressional District:**	2
Comments:	PI deceased.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	10/01/2004	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	4
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 09/30/2009 (from 3/31/2009) per NSSC info(04/2009) NOTE: Received NCE to 3/31/2009 (from 9/30/2008) per PI (9/08) NOTE: Funding goes through 9/30/2008 per D. Gruendel/KSC (9/07) NOTE: 8/2006--funding reinstated through 9/30/2007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kuang, Anxiu Ph.D. (University of Texas Pan American)		
Grant/Contract No.:	NNX07AT77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In our previous research on seed development in microgravity, changes were observed in storage reserve deposition during the late stages of seed development in the closed seed pods of Brassica rapa. Instead of storing protein and lipid, the spaceflight seeds retained starch as a major reserve, and protein bodies were small. These storage reserve changes led to decreased seed size and decreased vigor of second-generation plants. Vegetative development and earlier stages of reproductive development were unaffected by spaceflight, leading us to hypothesize that lack of convective air movement in the gaseous microenvironment inside the seed pod caused the change in storage reserved in microgravity. This hypothesis will be tested directly by comparing the gas composition of this small captive air space in 1-g and microgravity, using a rapid sampling procedure and post-flight analysis. Storage reserves in the developing seeds and growth of second generation seedlings and plants will be analyzed to assess the biological consequences of late stage seed development in microgravity. Because the experiment utilizes pre-formed seed pods in a tissue culture system, it is perfectly suited for an 11-day Shuttle mid-deck location in existing hardware like the Plant Growth Facility. Successful seed production under spaceflight conditions is central to NASA's goal of providing a renewable food source on future long-duration missions. Furthermore, a better understanding of the role of the seed microenvironment in determining storage reserves also has agricultural implications. NOTE: Project transferred to GROUND research program (FY2004)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A better understanding of the role of the seed microenvironment in determining storage reserves has important agricultural implications. These experiments will help clarify the relationship between rate of oxygen availability inside the seed pod and protein, lipid, and starch deposition in the seed. The basic information can be used in strategies to enhance storage reserves in seed crops here on earth. Further, the information gained on plant responses across the gravity continuum (micro-g through 4-g) give predictive insight for challenges plants may face when growing in fractional g-levels (such as on the Moon or Mars). Our three papers published in 2009 provide unique data in this area.
Task Progress:	Our project changed from the one initially described in the abstract due to changing programmatic needs at NASA following the award of NAG10-329. Initially the project was funded as a flight experiment. Following the Columbia disaster in February 2003, we received low level funding to continue pre-planning and methods development. By 2004 it was clear that there would be no opportunity to fly the experiment due to constraints on remaining Shuttle flights. The project was transferred to the ground-based research program. Included in the final report is a copy of the replanning request. Essentially we requested to pursue the same hypothesis that had been the basis of our flight experiment using the tool of hypergravity rather than microgravity. Thus in 2005 we began actively planning for a series of experiments on the large diameter centrifuges at the Center for Gravitational Biology Research facility at NASA Ames Research Center. Key to the success of our experiments was the willingness of NASA-Kennedy Space Center administrators to allow us to use flight hardware for our hypergravity experiments. Because of this decision, our team was able to use the Plant Growth Unit (used previously by our group for microgravity experiments on STS-54, STS-51, STS-68) and the Plant Growth Facility (used previously by our group for microgravity experiments on STS-87). Personnel at KSC worked with personnel at ARC to figure out how the powered hardware could be mounted to the large diameter rotors. The PGF was mounted on the "24-ft" rotor and the PGU was mounted on the "8-ft" rotor. The hardware was contained within powered swinging bucket cabinets on the rotors. In each case, a "ground control" (stationary 1-g) treatment was achieved in a second piece of hardware contained within a stationary cabinet providing the same environmental conditions as the rotor treatment. Experiments were conducted at 2-g and 4-g. Because of the exciting results obtained at 4-g, an additional experiment set was performed at 4-g. Funding and logistics constraints (closure of the CGBR) precluded additional experimentation on the large diameter rotors, but the PI was able to continue some laboratory-based experiments at hypergravity using a modified commercial benchtop centrifuge. Two plant species were used, along with the protocols used in prior microgravity experiments: Arabidopsis thaliana (STS-68; 11 days) and Brassica rapa (STS-87; 16 days). This strategy allowed us to compare a number of plant responses across a gravity range of micro-g through 4-g. These were the first experiments conducted of plant reproductive development at hypergravity. In both plants, flowering and early stages of seed development occurred during the 11 or 16 day experiments. Most significant for us was the opportunity to directly test the hypothesis that gravity controls the gaseous composition of the microenvironment around developing seeds, and that the changes observed in seed quality when seeds are produced in altered gravity is attributable to the change in the seed's microenvironment. Early in the grant cycle we developed a method for rapidly sampling the gaseous composition of the small captive airspace around developing seeds of Brassica (the seeds develop within a special hollow pod structure called a silique). This method involved in vitro culture of siliques: the detached pods are able to mature seeds when the pods tips are inserted into a culture medium and kept under aseptic conditions. This technique allowed us to produce a cohort of same-aged siliques, expose them to different hypergravity levels for different lengths of time, and then sample the enclosed gases. Our results of this methods development constituted a peer-reviewed publication (Musgrave et al., 2008; In vitro seed maturation in Brassica rapa L.: Relationship of silique atmosphere to storage reserve deposition. Environ. Exp. Bot. 62: 247-253.) Similarly, methods development for seed storage reserve analysis resulted in another publication (Musgrave et al., 2005. Seed storage reserves and glucosinolates in Brassica rapa L. grown on the International Space Station. J. Amer. Soc. Hort. Sci. 130(6): 848-856). This publication was the cover story of the journal for that month, and subsequently received the outstanding publication award from the American Society for Horticultural Science in 2006. Because of changes in grant management during the period of our project, analysis of the centrifuge experiments themselves was funded through a different contract (NNX07AT77G). The publication that represents the main test of our hypothesis recently appeared (Musgrave et al., 2009. Brassica rapa L. seed development in hypergravity. Seed Science Research 19: 63-72.) Here we report that hypergravity enhances gas availability to the developing embryos, resulting in more lipid and less starch and soluble carbohydrates. Although we had suspected this would be the case, it was the first time that changes in the gas composition of captive air surrounding seeds has been measured in altered gravity. Gravity's role in seed development is of importance to the space program because of the plan to use plants for food production and habitat regeneration in extraterrestrial settings.

Additional results from our experiments were peripheral to our central hypothesis, but no less significant scientifically. Using both Arabidopsis and Brassica we documented unexpected changes in plant growth form when these plants were grown in hypergravity during the flowering stage. At this part of the plants' life cycle, they undergo rapid changes in height as the flowering stalk appeared from the basal leaves and lengthens. With increasing g-load, the plants assume a more prostrate architecture. At 4-g, the larger Brassica plants grow almost parallel to the substrate. Concomitant with this change in architecture is a thickening of the stems and a decrease in production of the plant secondary compounds glucosinolates, thus altering the flavor and nutritional qualities of the plants. Because of prior data on these same parameters obtained in microgravity, we were able to examine how increasing g-force affects these basic components of plant growth and development. The full report was recently published (Allen et al., 2009. Gravity control of growth form in Brassica rapa and Arabidopsis thaliana (Brassicaceae): consequences for secondary metabolism. Amer. J. Bot. 96(3): 652-660.)

A very surprising result was our finding that seed production in Arabidopsis is sensitive to hypergravity. Although seed production occurred normally at 2-g, it was essentially absent at 4-g. Using the same battery of tests that we had employed during our microgravity experiments on STS-68, we determined that something was going wrong during the stage of pollen tube growth, preventing fertilization of the normally developed ovules. Subsequent experiments in our home laboratory focused on in vitro pollen germination assays at different g-levels. We show that Arabidopsis pollen tube growth is sensitive to g-level. Comparing the tube growth by Brassica pollen we saw that this species, which successfully produces seeds at 4-g, is also sensitive to hypergravity, however the threshold for damage occurs at a higher g-level than Arabidopsis. This is an important finding because it points to differential pollen tube responses to g in different plants, which could have consequences for plant fitness when being grown over multiple generations in non-earthnormal gravity. The full report was recently published (Musgrave et al., 2009. Hypergravity prevents seed production in Arabidopsis by disrupting pollen tube growth. Planta 230: 863-870.)

Finally, the loss of flight experiment opportunities, and indeed, plant gravitational biology research opportunities in general, during the period of this award spurred me to write a review article that contains the many things I have learned about growing plants in space during my years of funding from NASA. (Musgrave, 2007. Growing plants in space. CAB Reviews: Perspectives in Agriculture, Veterinary Science, Nutrition and Natural Resources doi: 10.1079/PAVSNNR20072065). The article was dedicated to the memory of Guy Etheridge, whose death in 2006 prevented him from seeing the results that his leadership as a NASA manager allowed us to deliver.

Bibliography Type:	Description: (Last Updated: 10/13/2009)
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